

Backscatter Rate Curvefit

In this section we present a power law fit for backscatter ratio as a function of loss cone filling derived from ELFIN data. The data we are fitting to is shown in Figure 1 (reproduced from manuscript). For this fit, we will only consider the case with low precipitating fluxes excluded, shown in the right panel.

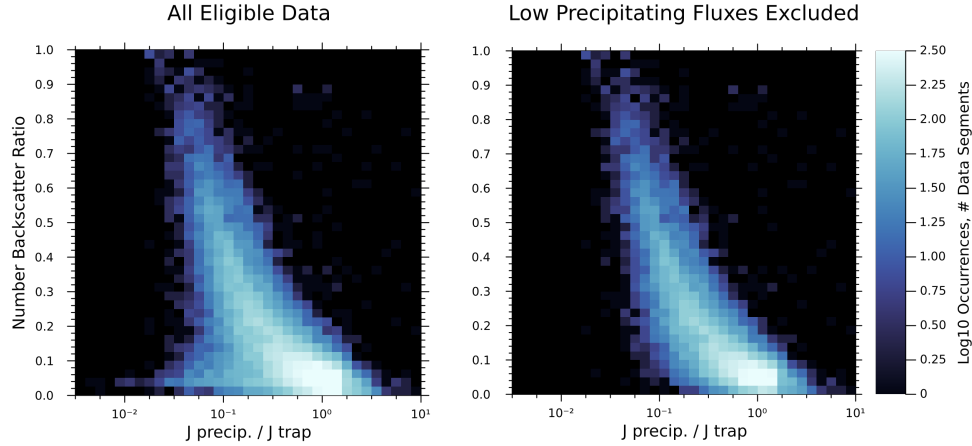


Figure 1: Number backscatter ratio vs. loss cone filling strength derived from 63 hours (69,510 spacecraft spins) of ELFIN data. Any measurement of backscatter ratio with absolute uncertainty greater than $\sigma \approx 0.025$ was discarded. Left: Results all eligible ELFIN data (see manuscript). Right: Measurements with precipitating flux less than $10^{2.75}$ electrons- $\text{cm}^{-2}\text{-s}^{-1}$ removed. Energy backscatter ratio has a similar pattern to number backscatter ratio and is not shown for brevity.

First we find the median backscatter ratio for every $J_{\text{precip}}/J_{\text{trap}}$ bin. This series, along with the 25th and 75th percentile of the r_N distribution for each $J_{\text{precip}}/J_{\text{trap}}$ bin, is shown in Figure 2 below.

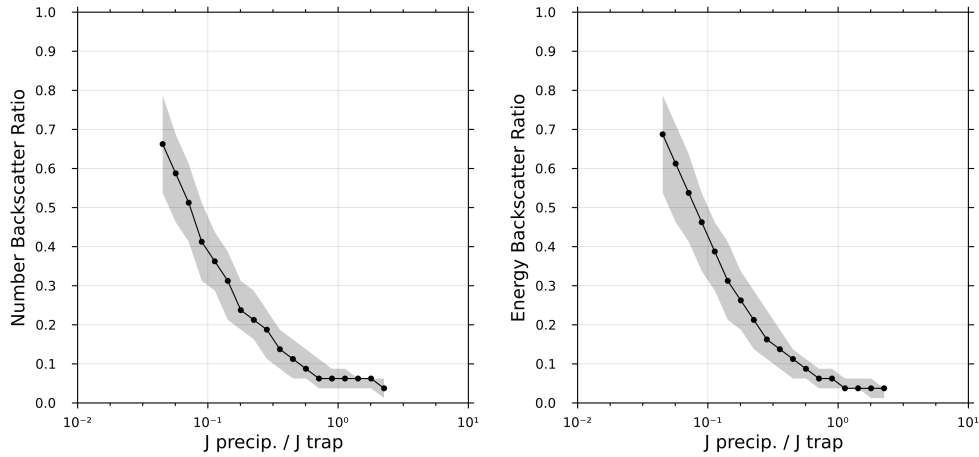


Figure 2: Median backscatter (black line) as well as the 25th and 75 percentile (grey ribbon below and above, respectively) as a function of precipitation strength. Left panel shows the number backscatter ratio r_N , right panel shows the energy backscatter r_E .

We then fit a power law to each curve using a least-squares regression. The formulae for these fits and their coefficient of determination R^2 are given below.

$$r_N = 0.068299421 \cdot \left(\frac{J_{\text{precip}}}{J_{\text{trap}}} \right)^{-0.73878583} \quad R^2 = 0.99336882$$

$$r_E = 0.05627575 \cdot \left(\frac{J_{\text{precip}}}{J_{\text{trap}}} \right)^{-0.84640575} \quad R^2 = 0.98450499$$

These fits are plotted over the data in Figure 3 below.

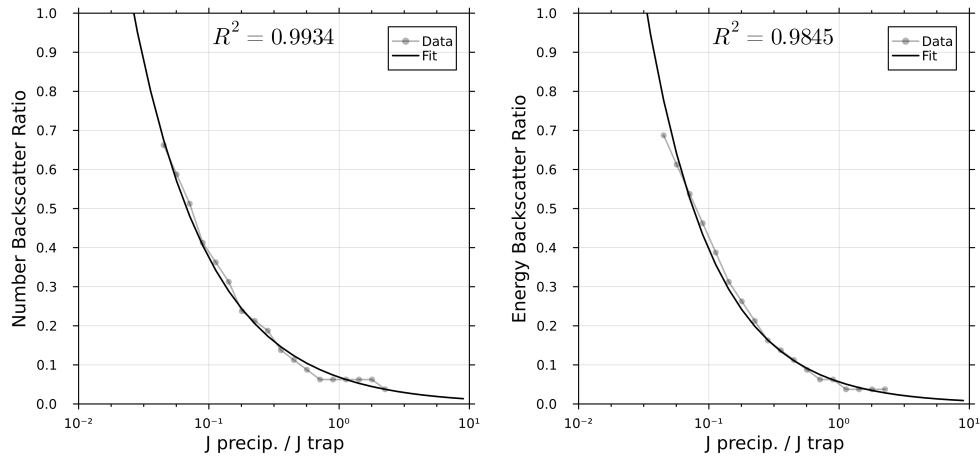


Figure 3: Median backscatter rates (grey line with dots) at a variety of loss cone filling levels with a power law curvefit (black line) overlaid. Left: Number backscatter ratio as a function of loss cone filling. Right: Energy backscatter ratio as a function of loss cone filling.

Energy Backscatter Residuals

ELFIN-derived energy backscatter plotted against simulation-predicted energy backscatter is shown below.

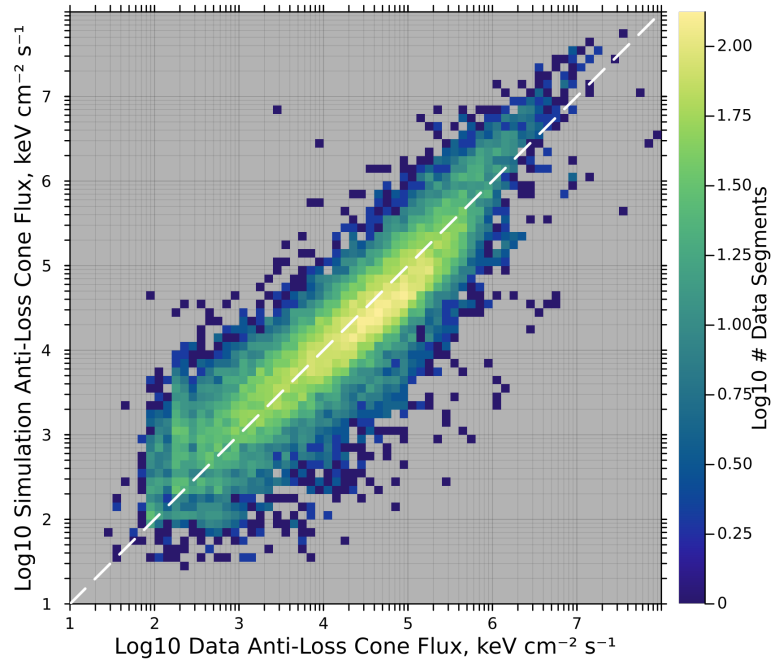


Figure 4: Anti-loss cone energy flux predicted by lookup tables derived from our Monte Carlo simulation vs. ELFIN-measured anti-loss cone energy for all ELFIN data segments that were simulated. The dashed white line indicates where predicted backscatter was equal to ELFIN-measured backscatter.